

Write legibly showing all the steps VERY clearly. Max. score=100 (Total points = 105).

- (1) (a) Let $(\Omega, \mathcal{F}, P) = ([0, 1], \mathcal{B}([0, 1]), m)$, where m denotes the Lebesgue measure on $[0, 1]$. Define the sequence of random variables X_n such that $X_1 \equiv 0$, $X_2 \equiv 13$ and $X_n(\omega) = n\mathbf{I}_{[\frac{1}{2}-\frac{1}{n}, \frac{1}{2}+\frac{1}{n}]}(\omega)$, $\omega \in \Omega$, $n \geq 3$. Let $X \equiv 0$.
- (i) Prove that $X_n \rightarrow X$ a.e. (P) as $n \rightarrow \infty$.
 - (ii) Prove: $\{X_n\}$ **not** uniformly integrable.
 - (iii) Prove: X_n does **not** converge to X in L^1 as $n \rightarrow \infty$.
- (b) In general, under what conditions on the functions $\{X_n\}$ would
 “ $X_n \rightarrow X$ a.e (P)” imply “ $X_n \rightarrow X$ in L^p ” for $p \geq 1$?
 ((3×8)+8=32 points)
- (2) Let $(\Omega, \mathcal{F})$ be a measurable space, and μ, μ_1, μ_2, ν are σ -finite measures on it. Define the set function: $\mu_3 \equiv 2\mu_1 + 3\mu_2$ (i.e., for all $A \in \mathcal{F}$, $\mu_3(A) = 2\mu_1(A) + 3\mu_2(A)$). Then, prove the following:
- (a) μ_3 is a measure.
 - (b) If $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\mu_3 \ll \nu$.
 - (c) Express the Radon-Nikodym derivative $\frac{d\mu_3}{d\nu}$ in terms of $\frac{d\mu_1}{d\nu}$ and $\frac{d\mu_2}{d\nu}$ (proof required).
 - (d) Let $\mu = \mu_3$ defined as defined above where $\mu_1 = \text{Normal}(2, 3)$, $\mu_2 = \text{Poisson}(6)$. Find the Lebesgue decomposition of μ with respect to $\nu = m$ = the Lebesgue measure on $\mathbb{R}$, and find the Radon Nikodym derivative $\frac{d\mu_a}{d\nu}$, where μ_a is the absolutely continuous component of μ in the decomposition.
 (4×10=40 points)
- (3) Suppose X is a random variable defined on some probability space $(\Omega, \mathcal{F}, P)$.
- (a) If $M(t) = E(e^{tX})$, $t \in \mathbb{R}$ is the moment generating function of X , then show
- $$P(X > a) \leq \inf_{t>0} e^{-ta} M(t), \quad a > 0.$$
- (b) If $X > 1$ a.e (P), then prove $E(\log X) \leq \log(EX)$. (8+10=18 points)
- (4) (a) State precisely the *Dominated convergence Theorem (DCT)* (Not the EDCT!).
- (b) Prove that the conclusion of DCT remains true even if “ $f_n \rightarrow f$ a.e(μ)” is replaced by “ $f_n \rightarrow f$ in measure” in the statement. (6 + 9 = 15 points)

TABLE 4.6.1. Some discrete univariate distributions.

Mean μ	$\mu(A), A \in \mathcal{B}(\mathbb{R})$
Bernoulli (p), $0 < p < 1$	$\sum_{i=0}^1 p^i (1-p)^{1-i} I_A(i)$
Binomial (n, p), $0 < p < 1, n \in \mathbb{N}$	$\sum_{i=0}^n \binom{n}{i} p^i (1-p)^{n-i} I_A(i)$
Geometric (p), $0 < p < 1$	$\sum_{i=0}^{\infty} p(1-p)^i I_A(i)$
Poisson (λ), $0 < \lambda < \infty$	$\sum_{i=0}^{\infty} e^{-\lambda} \frac{\lambda^i}{i!} \cdot I_A(i)$

TABLE 4.6.2. Some standard absolutely continuous distributions. Here m denotes the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Measure μ	$\mu(A), A \in \mathcal{B}(\mathbb{R})$
Uniform (a, b), $-\infty < a < b < \infty$	$m(A \cap [a, b]) / (b - a)$
Exponential (β), $\beta \in (0, \infty)$	$\int_A \frac{1}{\beta} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$
Gamma (α, β), $\alpha, \beta \in (0, \infty)$	$\int_A \frac{1}{\Gamma(\alpha) \beta^\alpha} x^{\alpha-1} \exp(-x/\beta) I_{(0, \infty)}(x) m(dx)$ where $\Gamma(a) = \int_0^\infty x^{a-1} e^{-x} dx, a \in (0, \infty)$
Beta (α, β), $\alpha, \beta \in (0, \infty)$	$\frac{1}{B(\alpha, \beta)} \int_A x^{\alpha-1} (1-x)^{\beta-1} I_{(0,1)}(x) m(dx),$ where $B(a, b) = \Gamma(a)\Gamma(b)/\Gamma(a+b), a, b \in (0, \infty)$
Cauchy (γ, σ) $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\pi \sigma} \frac{\sigma^2}{\sigma^2 + (x-\gamma)^2} m(dx)$
Normal (γ, σ^2), $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi\sigma}} \exp(-(x-\gamma)^2 / 2\sigma^2) m(dx)$
Lognormal (γ, σ^2), $\gamma \in \mathbb{R}, \sigma \in (0, \infty)$	$\int_A \frac{1}{\sqrt{2\pi\sigma}} \frac{e^{-(\log x - \gamma)^2 / 2\sigma^2}}{x} I_{(0, \infty)}(x) m(dx)$

1. (a) (i) For all $x \neq \frac{1}{2}$, $X_n(x) = 0$ for large n , and $P(\{\frac{1}{2}\}) = 0$. This proves $X_n \rightarrow X$ almost surely (P) as $n \rightarrow \infty$.
- (ii) Since for fixed $T > 0$, $\int |X_n| I_{\{|X_n| > T\}} dP = n \cdot m([\frac{1}{2} - \frac{1}{n}, \frac{1}{2} + \frac{1}{n}]) = n \cdot \frac{2}{n} = 2$, $\forall n > T$, $\{X_n\}$ is not UI.
- (iii) Here $E|X_n - X| = E(X_n) = n \cdot \frac{2}{n} = 2$ for all n large, so X_n does not converge to X in L^1 .
- (b) If $\{X_n^p\}$ is UI, and the measure is finite (which is the case in the example above), then $X_n \rightarrow X$ a.e (P) would imply $X_n \rightarrow X$ in L^p for $p \geq 1$ (HW problem: 2.36(b)). There are other correct answers too, for example, along the lines of DCT.

2. (a) $\mu_3(\emptyset) = 2\mu_1(\emptyset) + 3\mu_2(\emptyset) = 0$;
 $\mu_3(A) = 2\mu_1(A) + 3\mu_2(A) \geq 0 \Rightarrow \mu_3(A) \in [0, \infty]$, $\forall A \in \mathcal{F}$;
 If $\{A_i\}$ is a disjoint collection of sets in $\mathcal{F}$, then $\mu_3(\cup_i A_i) = 2\mu_1(\cup_i A_i) + 3\mu_2(\cup_i A_i) = 2 \sum_i \mu_1(A_i) + 3 \sum_i \mu_2(A_i) = \sum_i (2\mu_1(A_i) + 3\mu_2(A_i)) = \sum_i \mu_3(A_i)$.
 So μ_3 is a measure.
- (b) $\nu(A) = 0 \Rightarrow \mu_1(A) = \mu_2(A) = 0 \Rightarrow \mu_3(A) = 2\mu_1(A) + 3\mu_2(A) = 0$. So $\mu_3 \ll \nu$.
- (c) Claim: $\frac{d\mu_3}{d\nu} = 2\frac{d\mu_1}{d\nu} + 3\frac{d\mu_2}{d\nu}$ a.e (ν). *Proof:* Since $\mu_1 \ll \nu$ and $\mu_2 \ll \nu$, then $\exists f_1, f_2 \geq 0$ s.t. $\mu_1(A) = \int_A f_1 d\nu$, $\mu_2(A) = \int_A f_2 d\nu$. By definition, $\mu_3(A) = 2\mu_1(A) + 3\mu_2(A) = 2 \int_A f_1 d\nu + 3 \int_A f_2 d\nu = \int_A (2f_1 + 3f_2) d\nu$ by linearity of integrals. Hence by uniqueness of RN derivatives, $\frac{d\mu_3}{d\nu} = 2f_1 + 3f_2 = 2\frac{d\mu_1}{d\nu} + 3\frac{d\mu_2}{d\nu}$ a.e (ν).
- (d) Let $B = \mathbb{N} \cup \{0\} = \mathbb{N}_0$, then $3\mu_2(B^c) = 0 = \nu(B)$, i.e. $3\mu_2 \perp \nu$. Choose $\mu_s = 3\mu_2$, $\mu_a = 2\mu_1$, then $\mu = \mu_a + \mu_s$. To show $\mu_a \ll \nu$ and compute $\frac{d\mu_a}{d\nu}$.
 Note that if $\phi_{2,3}$ represent the density of $Normal(2, 3)$, then by definition $\mu_1(A) = \int_A \phi_{2,3} dm$ and $\nu = m$. So, $\frac{d\mu_1}{d\nu} = \phi_{2,3}$ from RN theorem. Hence, from part (a) above, $\frac{d\mu_a}{d\nu} = \frac{d(2\mu_1)}{d\nu} = 2\frac{d\mu_1}{d\nu} = 2\phi_{2,3}$.

3. (a) Using Markov's inequality, for each $t > 0$,

$$P(X > a) = P(e^{tX} > e^{ta}) \leq e^{-ta} M(t),$$

and hence the result follows.

- (b) Let $\phi(x) = -\log x$, $x > 1$, then $\phi(x)$ is a convex function.
 If $EX < \infty$, then since $0 \leq \log x \leq x$, $\forall x \geq 1$, we have $0 \leq E(\log X) \leq EX$, i.e. $E \log X < \infty$. By Jensen's inequality, $E\phi(X) \geq \phi(EX)$, i.e. $E(\log X) \leq \log(EX)$.
 If $EX = \infty$, then $E \log X \leq \infty = \log(EX)$ holds trivially.
 Hence $E(\log X) \leq \log(EX)$.

4. (a) Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and f_n, f, g be measurable functions. If $|f_n| \leq g, a.e. (\mu), \forall n \geq 1, \int g d\mu < \infty$ and $f_n \rightarrow f a.e. (\mu)$, then $f \in L^1(\Omega, \mathcal{F}, \mu), \lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$ and $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.

(b) We need to prove the following modified DCT:

Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and f_n, f, g be measurable functions. If $|f_n| \leq g, a.e. (\mu), \forall n \geq 1, \int g d\mu < \infty$ and $f_n \rightarrow f$ in measure, then $f \in L^1(\Omega, \mathcal{F}, \mu), \lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$ and $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.

Proof: First we state two facts in real analysis:

Fact 1: A sequence $\{x_n\}$ converges to x iff for any subsequence $\{x_{n_k}\}$ of $\{x_n\}$, there is a further subsequence $\{x_{n_{k_l}}\}$ of $\{x_{n_k}\}$ such that $x_{n_{k_l}} \rightarrow x$.

Fact 2 (Theorem 2.5.2): $f_n \rightarrow f$ in measure $\Rightarrow$ there exists a subsequence $\{f_{n_k}\}$ such that $f_{n_k} \rightarrow f a.e. (\mu)$.

Then we prove $f \in L^1(\Omega, \mathcal{F}, \mu)$ and $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$.

Denote $a_n = \int f_n d\mu, a = \int f d\mu$ and $b_n = \int |f_n - f| d\mu$. To show $a_n \rightarrow a$ and $b_n \rightarrow 0$, we need to use Fact 1. Choose any subsequence $\{a_{n_k}\}$ of $\{a_n\}$, and for this subsequence, choose the corresponding subsequence functions $\{f_{n_k}\}$ of $\{f_n\}$. Since $f_n \rightarrow f$ in measure, then $\lim_{n \rightarrow \infty} \mu(|f_n - f| > \varepsilon) = 0, \forall \varepsilon > 0$, thus $\lim_{k \rightarrow \infty} \mu(|f_{n_k} - f| > \varepsilon) = 0, \forall \varepsilon > 0$, i.e. $f_{n_k} \rightarrow f$ in measure. By Fact 2, we can get a sub-subsequence $f_{n_{k_l}} \rightarrow f a.e. (\mu)$. For this sub-subsequence $\{f_{n_{k_l}}\}$, conditions of DCT are true. Hence from DCT, we have $f \in L^1(\Omega, \mathcal{F}, \mu)$ and $\lim_{l \rightarrow \infty} \int f_{n_{k_l}} d\mu = \int f d\mu$, i.e. $a_{n_{k_l}} \rightarrow a$. By Fact 1, we get $a_n \rightarrow a$, i.e. $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$. Similarly, we have $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.

Midterm 2 (2015)

① (a) (i) We want to prove that $X_n \rightarrow X$ a.e. (P) as $n \rightarrow \infty$.

Consider that

$$X_n(\tfrac{1}{2}) \rightarrow \infty \text{ as } n \rightarrow \infty$$

but

$$X_n(w) \rightarrow 0 \text{ as } n \rightarrow \infty \text{ for all } w \in [0,1] \text{ s.t. } w \neq \tfrac{1}{2}.$$

Furthermore, $P(\{\tfrac{1}{2}\}) = 0$ (since Lebesgue measure).

Thus, for $B = \{\tfrac{1}{2}\}$ with $P(B) = 0$,

$$\lim_{n \rightarrow \infty} X_n(w) = 0 \quad \forall w \in B^c.$$

Hence, $X_n \rightarrow X$ a.e. (P) as $n \rightarrow \infty$.

(ii) In order for $\{X_n\}$ to be UI, it would need to be the case that

$$\sup_{n \in \mathbb{N}} a_n(t) \rightarrow 0 \text{ as } t \rightarrow \infty$$

where

$$a_n(t) = \int |X_n| \cdot \mathbb{I}\{X_n > t\} dP.$$

Choose some $T > 0$. Then for $n > T$

$$a_n(T) = \int |X_n| \cdot \mathbb{I}\{X_n > T\} dP$$

$$= \int |n \mathbb{I}[\tfrac{1}{2} - \tfrac{1}{n}, \tfrac{1}{2} + \tfrac{1}{n}]| dP$$

$$= \int n \mathbb{I}[\tfrac{1}{2} - \tfrac{1}{n}, \tfrac{1}{2} + \tfrac{1}{n}] dP$$

$$= n P([\tfrac{1}{2} - \tfrac{1}{n}, \tfrac{1}{2} + \tfrac{1}{n}]) = n \left(\tfrac{1}{2} + \tfrac{1}{n} - \tfrac{1}{2} + \tfrac{1}{n} \right) = 2$$

Thus, for all $n > T$,

$$a_n(T) = 2.$$

Since T was arbitrary, this is true for all T .

Then

$$\sup_{n \in \mathbb{N}} a_n(t) = 2 \quad \forall t, \text{ so}$$

$$\sup_{n \in \mathbb{N}} a_n(t) \not\rightarrow 0 \text{ as } t \rightarrow \infty.$$

Hence, $\{X_n\}$ is not UI.

(iii) Now want to prove that X_n does not converge in L^1 as $n \rightarrow \infty$. That is, we want to show that

$$\lim_{n \rightarrow \infty} \int |X_n - X| dP \neq 0.$$

Consider the following...

$$\lim_{n \rightarrow \infty} \int |X_n - X| dP = \lim_{n \rightarrow \infty} \int |n I[\frac{1}{2} - \frac{1}{n}, \frac{1}{2} + \frac{1}{n}] - 0| dP$$

$$= \lim_{n \rightarrow \infty} \int n I[\frac{1}{2} - \frac{1}{n}, \frac{1}{2} + \frac{1}{n}] dP$$

$$= \lim_{n \rightarrow \infty} n P(\frac{1}{2} - \frac{1}{n}, \frac{1}{2} + \frac{1}{n})$$

$$= \lim_{n \rightarrow \infty} n \left(\frac{1}{2} + \frac{1}{n} - \frac{1}{2} + \frac{1}{n} \right)$$

$$= \lim_{n \rightarrow \infty} 2$$

$$= 2 \neq 0.$$

Hence $X_n \not\stackrel{L^1}{\rightarrow} X$. ✓

(b) The functions $\{X_n\}$ would need to have the conditions of

• $\{X_n\}_{n=1} \subset L^p(\Omega, \mathcal{F}, P)$

• $\mu(\Omega) < \infty$

• $\{X_n^p\}_{n=1}$ is UI

in order for $X_n \rightarrow X$ a.e.(P) to imply $X_n \xrightarrow{L^p} X$ $p \geq 1$.

(2) $(\Omega, \mathcal{F})$ a measurable space and μ, μ_1, μ_2, ν σ -finite measures on $(\Omega, \mathcal{F})$.

$$\mu_3 = 2\mu_1 + 3\mu_2$$

(a) WTS μ_3 a measure

(i) $\mu_3(\emptyset) = 2\mu_1(\emptyset) + 3\mu_2(\emptyset) = 2(0) + 3(0) = 0$ ✓

(ii) $\mu_1(A) \in [0, \infty)$ & $\mu_2(A) \in [0, \infty)$ $\forall A \in \mathcal{F}$

Thus,

$$\mu_3(A) = 2\mu_1(A) + 3\mu_2(A) \in [0, \infty). \quad \checkmark$$

(iii) Let $A_1, A_2, \dots \in \mathcal{F}$ be disjoint sets and $\bigcup_{n=1} A_n \in \mathcal{F}$ since $\mathcal{F}$ a σ -algebra. Then

$$\begin{aligned} \mu_3\left(\bigcup_{n=1} A_n\right) &= 2\mu_1\left(\bigcup_{n=1} A_n\right) + 3\mu_2\left(\bigcup_{n=1} A_n\right) \\ &= 2\sum \mu_1(A_n) + 3\sum \mu_2(A_n) \\ &= \sum 2\mu_1(A_n) + 3\mu_2(A_n) = \sum \mu_3(A_n) \quad \checkmark \end{aligned}$$

Hence, μ_3 is a measure.

(c) Want to express the R-N derivative $\frac{d\mu_3}{d\nu}$ in terms of $\frac{d\mu_1}{d\nu}$ and $\frac{d\mu_2}{d\nu}$.

Since $\mu_1 \ll \nu$, $\exists$ some non-negative measurable function h_1 on $(\Omega, \mathcal{F})$ s.t.

$$\mu_1(A) = \int_A h_1 d\nu \quad \forall A \in \mathcal{F},$$

and $\mu_2 \ll \nu \Rightarrow \exists$ some non-neg. meas. function h_2 on $(\Omega, \mathcal{F})$ s.t.

$$\mu_2(A) = \int_A h_2 d\nu \quad \forall A \in \mathcal{F}.$$

Then

$$\begin{aligned} \mu_3(A) &= 2\mu_1(A) + 3\mu_2(A) \\ &= 2 \int_A h_1 d\nu + 3 \int_A h_2 d\nu \\ &= \int_A (2h_1 + 3h_2) d\nu \quad (\text{by linearity of integrals}) \\ &= \int_A h_3 d\nu \end{aligned}$$

where

$$h_3 = \frac{d\mu_3}{d\nu} = 2 \frac{d\mu_1}{d\nu} + 3 \frac{d\mu_2}{d\nu} \quad \text{a.e. } (\nu)$$

which is unique by the uniqueness of the Radon-Nikodym derivative.

(d) Let $\mu = \mu_3$, $\mu_1 = \text{Normal}(2, 3)$, $\mu_2 = \text{Poisson}(6)$,
 where $\mu = 2\mu_1 + 3\mu_2$. Let $\nu = m = \text{Lebesgue}$
 measure. Then

$$\mu(A) = 2 \int_A f_1 dm + 3 \int_A f_2 d\mu_c \quad \swarrow \text{counting measure}$$

where

$$f_1 = \frac{1}{\sqrt{2\pi}(3)} \exp \left\{ -\frac{(x-2)^2}{2(3)^2} \right\}$$

$$f_2 = \frac{e^{-6} 6^i}{i!} \mathbb{I}_A(i) \quad \text{where } i \in \{0, 1, \dots\}.$$

Then let

$$\begin{aligned} \mu_a &= 2\mu_1 \\ \mu_s &= 3\mu_2 \end{aligned} \quad \text{where } \mu_a + \mu_s = 2\mu_1 + 3\mu_2 = \mu.$$

Then since we can write

$$\mu_a(A) = 2\mu_1(A) = \int_A 2f_1 dm$$

where $2f_1$ is a non-negative, measurable function, then
 $\mu_a \ll m$.

Let $B = \mathbb{N} \cup \{0\}$. Then

$$3\mu_2(B^c) = 3\mu_2((\mathbb{N} \cup \{0\})^c) = 0 = \nu(\mathbb{N} \cup \{0\}) = \nu(B).$$

Thus, $\mu_s \perp \nu$. Hence, the decomposition of
 μ into $\mu_a + \mu_s$ is the Lebesgue decomposition
 of μ wrt ν .

Now, we want to find the R-N derivative.

$$\frac{d\mu_a}{dv} = \frac{dZ\mu}{dv}.$$

By the R-N theorem, we have that

$$\frac{d\mu_a}{dv} = Zf_1. \quad \checkmark$$

③ X a r.v. on $(\Omega, \mathcal{F}, P)$

(a) Let $M(t) = E[e^{tx}] \quad t \in \mathbb{R}$.

$$\begin{aligned} P(X > a) &= P(e^{tx} > e^{ta}) \leq \frac{\int e^{tx} d\mu}{e^{ta}} \quad \forall t \\ &= E[e^{tx}] \cdot e^{-ta} \\ &= M(t) e^{-ta} \end{aligned}$$

$$\text{Thus, } P(X > a) \leq \inf_{t > 0} M(t) e^{-ta}. \quad \checkmark$$

(b) Let $X > 1$ a.e. (P) , $(P(X > 1) = 1)$

Consider that $\phi(x) = -\log(x)$ is a convex function for $x > 1$. Then by Jensen's inequality

$$\begin{aligned} \phi(E[X]) &\leq E[\phi(X)] \\ \Rightarrow -\log(E[X]) &\leq E[-\log(X)] \\ \Rightarrow \log(E[X]) &\geq E[\log(X)] \end{aligned}$$

(when $E[|X|] < \infty$)

(when $E[|X|] = \infty$,

then

$$\begin{aligned} \log(E[X]) &= \log(\infty) \\ &= \infty \\ &\geq E[\log(X)] \quad \checkmark \end{aligned}$$

④ (a) Let $f, \{f_n\}_{n=1}^\infty$, and g be measurable functions on $(\Omega, \mathcal{F}, \mu)$. If

(i) $|f_n| \leq g$ a.e. $(\mu) \quad \forall n \geq 1$,

(ii) $\int g d\mu < \infty$, and

(iii) $f_n \rightarrow f$ a.e. (μ) ,

then

(1) $f \in L^1(\Omega, \mathcal{F}, \mu)$,

(2) $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$, and

(3) $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.

● Want to prove that if $f, \{f_n\}_{n=1}^\infty, g$ are measurable functions on $(\Omega, \mathcal{F}, \mu)$, and if

(i) $|f_n| \leq g$ a.e. $(\mu) \quad \forall n \geq 1$,

(ii) $\int g d\mu < \infty$, and

(iii) $f_n \xrightarrow{m} f$,

then

(1) $f \in L^1(\Omega, \mathcal{F}, \mu)$,

(2) $\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$, and

● (3) $\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$.

Let $a_n = \int f_n d\mu$ and $a = \int f d\mu$. We want to show that $a_n \rightarrow a$. For any subsequence $\{a_{n_k}\}_{k=1}^\infty$,

consider the sequence $\{f_{n_k}\}_{k \geq 1}$ attached to it...

We know that

$$f_n \xrightarrow{m} f$$

$$\Rightarrow \lim_{n \rightarrow \infty} \mu(\{|f_n - f| > \varepsilon\}) = 0 \quad \forall \varepsilon > 0$$

$$\Rightarrow \lim_{k \rightarrow \infty} \mu(\{|f_{n_k} - f| > \varepsilon\}) = 0 \quad \forall \varepsilon > 0$$

$$\Rightarrow f_{n_k} \xrightarrow{m} f.$$

Then, this implies that there exists a sub-subsequence $\{f_{n_{k_j}}\}_{j \geq 1}$ s.t.

$$f_{n_{k_j}} \rightarrow f \text{ a.e. } (\mu).$$

Then, this subsequence meets all of the conditions, for the DCT. Thus,

$$(1) f \in L^1(\Omega, \mathcal{F}, \mu), \quad \checkmark$$

$$(2) \lim_{j \rightarrow \infty} \int f_{n_{k_j}} d\mu = \int f d\mu, \text{ and}$$

$$(3) \lim_{j \rightarrow \infty} \int |f_{n_{k_j}} - f| d\mu = 0.$$

Thus, the 1st part of the modified DCT has been proved. Then consider that there is a real analysis fact that says that a sequence

$\{a_n\}_{n \geq 1}$ converges to a iff $\forall \{a_{n_k}\}_{k \geq 1} \exists$ some $\{a_{n_{k_j}}\}_{j \geq 1}$ s.t. $a_{n_{k_j}} \rightarrow a$ as $j \rightarrow \infty$.

We have shown that for any subsequence
 $a_{n_k} = \int f_{n_k} d\mu$, $\exists$ some $a_{n_{k_j}} = \int f_{n_{k_j}} d\mu$
s.t.

$$\lim_{j \rightarrow \infty} \int f_{n_{k_j}} d\mu = \int f d\mu.$$

Hence, $(a_n \rightarrow a)$

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu. \quad \checkmark$$

Thus, the second result of the modified DCT
has been proved. Finally, let

$$b_n = \int |f_n - f| d\mu.$$

We want to show that $b_n \rightarrow 0$ as $n \rightarrow \infty$.
For any $\{b_{n_k}\}_{k \geq 1}$, $\exists$ some $\{b_{n_{k_j}}\}_{j \geq 1}$ s.t.

$b_{n_{k_j}} \rightarrow 0$ ($\lim_{j \rightarrow \infty} \int |f_{n_{k_j}} - f| d\mu = 0$). (We just
showed this.) Thus, we can apply the real
analysis fact again, to say that

$$b_n \rightarrow 0 \text{ as } n \rightarrow \infty$$

$$\Rightarrow \lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0. \quad \checkmark$$

